

# Supplementary Information

Repeatable station terms modify fixed-probability long-period response spectra at stations across Japan

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## Supplementary Methods

### Record selection and direct periods

The public J-SHIS/NIED `sub1-v2024` flatfile contains 333,808 K-NET and KiK-net records[1, 2]. The download page defines the subset by magnitude of at least 5 and shortest fault distance no greater than 300 km. All records in the archive have JMA magnitude  $M_{\text{JMA}} \geq 5$ ; F-net  $M_w$  is stored separately. Its site table codes ground-surface installations as 1 and borehole installations as 2. We retain code 1 for the residual model and response-spectrum propagation. Of 231,380 surface records, 8,675 lack finite F-net  $M_w$  and 41 additional records have nonpositive AVS30. The remaining 222,664 records have positive distance, supported MF2013 source classes and finite model predictions at all eight periods.

MF2013 uses RotD100, the maximum norm of two horizontal oscillator responses[3]. RotD100 spectral acceleration is read directly at 0.1, 0.2, 0.3, 0.5, 1.0, 2.0, 3.0 and 5.0 s. Every retained record has a positive value at each period, so the final count remains 222,664 throughout. No response coordinate is copied or interpolated.

RotD50 residuals are decomposed on the identical sample as a response-component sensitivity. The RotD50 and RotD100 fields are compared station by station, and record-level RotD50/RotD100 ratios quantify the change in absolute coordinate. Zhao 2006 uses RotD50 because OpenQuake defines the model for the geometric mean of two horizontal components and the flatfile does not provide that exact response coordinate.

### Residual model boundary

The primary prediction contains the MF2013 magnitude-distance backbone and the published D1400 and AVS30 terms[4, 5]. D1400 is the depth to the 1.4 km s<sup>-1</sup> shear-wave velocity horizon. AVS30 is the average shear-wave velocity in the upper 30 m. The anomalous-intensity sensitivity uses the public northeastern and southwestern geographic and depth rules[6]. Incomplete plate-membership metadata restrict this term to a sensitivity calculation. The Philippine Sea Plate term is constant across stations for a qualifying event at a given period and is absorbed by the event fixed effect. The fitted global intercept is retained as a model offset; formal assessment of MF2013 calibration lies outside this design.

### Fixed effect decomposition

For residual  $r_{ij}$  at station  $i$  during event  $j$ , the additive model is

$$r_{ij} = \mu + \delta E_j + \delta S_i + \epsilon_{ij}. \quad (1)$$

Here,  $\mu$  is the global intercept,  $\delta E_j$  is the between-event residual,  $\delta S_i$  is the site-to-site residual, and  $\epsilon_{ij}$  is the remaining within-site residual. We refer to  $\delta S_i$  as the station term. It can contain repeatable local site response, unresolved average path effects and persistent instrument effects. Alternating grouped least squares estimates  $\delta E_j$  and  $\delta S_i$ . Record-count-weighted event and station means are constrained to zero, so  $\mu$  contains the global offset. Iteration stops when the largest parameter change falls below  $10^{-10} \log_{10}$  units. All 24 period-model combinations converge; the largest final change is  $9.99 \times 10^{-11}$  and the largest iteration count is 1,161.

## Event group repeatability

The 1,737 usable earthquakes are assigned to five groups with a fixed random seed. The same partition is used at every period and for both ground-motion models. Each fold estimates station terms separately from four earthquake groups and the fifth group. The two station-term vectors are centred with their respective record counts. A station requires at least 15 records in the four-group estimate and five records in the independent fifth-group estimate. We report their correlation and the record-weighted root mean squared error of the fifth-group terms relative to a zero station term. Each fifth group contains 347 or 348 earthquakes, and every decomposition converges below  $10^{-10} \log_{10}$  units.

## Spatial estimation of station terms

Five spatial blocks are produced by  $k$ -means clustering of standardised longitude and latitude. VS10, VS20 and VS30 are borehole-profile averages of shear-wave velocity in the upper 10, 20 and 30 m. AVS30 is the corresponding upper-30-m average from the J-SHIS shallow-subsurface model. D1100, D1400, D1700 and D2100 are depths to the tops of layers with shear-wave velocities of 1.1, 1.4, 1.7 and 2.1 km s<sup>-1</sup>, and Dbase is the depth to seismological bedrock. Velocities are in m s<sup>-1</sup> and depths are in m. The site-parameter model uses these variables, elevation, sensor depth and network. The site-and-regional-parameter model adds longitude, latitude and distance from each station to the northeastern and southwestern Japanese volcanic fronts. Both histogram gradient-boosting models use fixed hyperparameters: learning rate 0.04, 400 iterations, 15 maximum leaf nodes, 35 minimum samples per leaf and L2 regularisation 0.1. Numeric variables are median-imputed within each estimation fold; network is mode-imputed and one-hot encoded. Estimates are generated only for the test block and are centred to a record-weighted zero mean before spectrum propagation. A separate complexity sensitivity changes maximum leaf nodes to 7 or 31, minimum samples per leaf to 70, L2 regularisation to 1.0, or the iteration budget to 200. Each setting retains the same blocks, preprocessing, weights and random seed.

## MF2013 applicability sensitivity

The MF2013 regression analysis selected  $M_w \geq 5.5$  ground-surface records with two horizontal components, at least five triggered stations per event and source distance below 200 km[4]. It also truncated records beyond the magnitude-dependent distance where the Kanno et al. median PGA fell below 10 cm s<sup>-2</sup>[7]. We reconstruct the base-model equations

$$\log_{10} PGA = aM_w + bX - \log_{10} (X + d10^{0.5M_w}) + c, \quad (2)$$

with PGA in cm s<sup>-2</sup> and source distance  $X$  in km. The coefficients are

$$\begin{aligned} (a, b, c, d) &= (0.556, -0.003070, 0.2560, 0.00547), & H \leq 30 \text{ km}, \\ (a, b, c, d) &= (0.409, -0.00389, 1.5600, 0), & H > 30 \text{ km}. \end{aligned}$$

The Kanno station-amplification term is excluded because MF2013 applies this screen as a magnitude-distance boundary. The relation is evaluated at each record distance after the  $M_w$  and 200 km screens. Events with fewer than five remaining stations are then removed. The resulting sample contains 35,857 records from 411 earthquakes, including 578 stations with at least 20 records. Residual decomposition, independent-earthquake-group comparison, spatial estimation and spectrum propagation are repeated without changing the remaining settings.

## KiK-net paired-sensor validation

Ground-surface and borehole rows at a KiK-net site are paired by earthquake and site code. The pairing is one-to-one and yields 102,428 positive RotD100 pairs at every analysed period, spanning 699 sites and 1,410 earthquakes. For each station and period, we calculate the median  $\log_{10}$  ratio of surface to borehole RotD100 spectral acceleration. Pearson, Spearman and record-count-weighted correlations compare this response-

spectrum ratio with the primary surface station term. A depth-adjusted Pearson correlation first removes separate linear trends with log borehole depth from both quantities.

The primary five-group earthquake partition supplies an independent-group test. For each fold, station terms are estimated separately from four earthquake groups and the fifth group. A weighted linear relation between response-spectrum ratios and surface station terms is fitted with the four groups, then applied to station terms estimated from the fifth group. The fifth-group estimate uses no response-spectrum ratio from those earthquakes. Stations require at least 15 pairs in the four groups and five surface records in the fifth group. Ratio repeatability compares medians formed independently from the two earthquake sets.

The spectral-shape calculation retains stations represented at all eight periods. Separate station and period means are removed from the station-term and surface-to-borehole response-spectrum-ratio matrices. This isolates station-specific period dependence from average amplification and the network-wide response curve. Cluster-bootstrap intervals resample complete station spectra. In each fold, ratio spectra use four earthquake groups and station-term spectra use the fifth; two-way centring is repeated within the fold.

## Alternative ground-motion-model calculation

We evaluate the Zhao et al. 2006 active-crustal, interface and intraslab equations with OpenQuake Engine 3.25.1[8, 9]. The flatfile source class selects the equation branch. Moment magnitude, hypocentral depth and crustal rake are read from the source table; documented shortest fault distance is supplied as rupture distance, and AVS30 is supplied as  $V_{S30}$ . RotD50 is used as the closest orientation-independent coordinate to the model’s geometric-mean target. The same surface observations, periods, station threshold, earthquake groups and spatial blocks are used for the MF2013 and Zhao calculations. This changes the regional ground-motion model while retaining the data and evaluation design.

## European external replication

Version 3 of the Engineering Strong-Motion database was queried through its flatfile web service[10]. The local snapshot contains 41,841 rows and events through 7 July 2026; its SHA-256 digest is 084ef4c7519fdeb26a05db9ad980535cd4c2207b2358d6b668cd5dafb4f0c724. The query requests BEST and GOOD HN, HG and HL records with magnitude at least 4.5, epicentral distance no greater than 300 km, no late trigger and spectra at 1.0, 2.0 and 3.0 s. The primary sample retains  $4.5 \leq M_w \leq 7.6$  shallow events and stations within  $25\text{--}72^\circ$  N and  $25^\circ$  W– $45^\circ$  E, surface instruments away from structures, positive VS30 and positive as-recorded horizontal spectra. Events begin on 1 January 2015, after publication of the fixed Bindi et al. model[11]. Multiple channels for one event–station pair are ranked by record quality, manual processing and channel type before deduplication. Iterative support filtering requires five records per event and three per station, leaving 13,430 records from 634 events and 539 stations.

The primary residual subtracts the OpenQuake Bindi et al. hypocentral-distance model with Eurocode 8 site classes and unspecified faulting from the geometric mean of the as-recorded horizontal spectra. Disconnected station–event graph components are solved separately. Five fixed earthquake groups are decomposed independently, with at least 15 records in the four-group estimate and five records in the fifth-group estimate at each station. The two vectors are centred within their intersecting graph-component pair, using fifth-group record counts. Only component pairs with ten stations enter the pooled statistic. Station-cluster bootstrap intervals use 2,000 replicates. A magnitude sensitivity raises the lower bound to  $M_w = 5.0$ . A second sensitivity uses the Kotha et al. model, its  $M_w \leq 7.4$  regression range, Joyner–Boore distance and RotD50[12]; its smaller connected sample uses thresholds of ten records in the four groups, three records in the fifth group and three stations per component pair.

For cross-region estimation, the fixed Japanese K-NET relation retains  $\log_{10}(V_{S30})$  and elevation, the only continuous site variables available in both archives. It is applied without refitting to the 265 ESM stations with at least 15 primary records. The record-weighted mean of the estimates is removed without using ESM station terms. Sedimentary-depth and volcanic-front variables are omitted because ESM does not supply harmonised counterparts.

Local ESM spatial estimation is evaluated in the largest connected station–event component, which con-

tains 236 stations with at least 15 records. We request five  $k$ -means blocks and reduce the count until every block contains at least 30 stations; the rule selects three blocks without access to response values. The common-site model uses  $\log_{10}(V_{S30})$  and elevation. The metadata model adds slope and the ESM housing, installation and reference-site categories. The metadata-and-location model also adds longitude and latitude. Median imputation, fold-specific categorical encoding, model hyperparameters, station weights and 2,000-replicate station bootstrap follow the Japanese spatial test. Categorical fields with no observed estimation value are omitted in that fold. A second design separates both one earthquake group and one spatial block: station terms from four earthquake groups fit the model outside the block, and independently estimated fifth-group terms inside the block provide the score. No result is used for hyperparameter or block selection.

## Cross-network and event-influence tests

The first cross-network test estimates a relation with K-NET and applies it to KiK-net, then reverses the direction. A common-variable model removes VS20, VS30 and network identity because these fields have different completeness in the two networks. It retains VS10, AVS30, sedimentary depths, elevation, sensor depth, coordinates and volcanic-front distances.

A stricter test separates both network and earthquake sample. We retain 1,286 earthquakes with at least five records in each network and assign them to five fixed groups. K-NET station terms from four groups fit models with the fixed primary settings; KiK-net station terms are estimated only from the fifth group. The site-parameter model contains no network indicator. Source stations require 15 records in the four groups and target stations require five records in the fifth group. Estimates are centred by the mean source-network estimate without target recentering. The reverse direction uses the same groups and settings. Target-station bootstrap intervals use 2,000 replicates.

The chronological test retains 1,277 of these earthquakes with finite origin times and orders them before any station-term estimation. Fixed early/late splits of 70/30, 80/20 and 90/10 are evaluated. Early-earthquake station terms from one network fit a fixed relation, and later-earthquake station terms from the other network supply the comparison. The variables, thresholds, model settings and source-only centring match the random-group test. Origin time is used only to assign earthquakes. Every cutoff and both network directions are retained in the output. An earthquake-level split table records each event, timestamp and partition.

Fixed macroregions are defined by reproducible latitude and longitude boundaries. Event influence is measured after deleting each of the ten events with the largest 3.0 s record count and after deleting all ten at every period. A record-weighted mean shift aligns each perturbed set of station terms with the full-sample terms before comparison.

## Path and source stratification

Records are grouped into four hypocentral-bearing sectors, four shortest-fault-distance bins and three source classes. Bearing is the initial great-circle direction from the source horizontal coordinates to the station. Stratification can disconnect the bipartite station-event graph, especially within 50 km. Each graph component is solved separately by sparse least squares. Stations require ten records within a stratum, retained components require ten supported stations, and every component is aligned to the full-sample station terms by its record-weighted mean difference. Across all 88 period-stratum fits, the maximum relative normal-equation residual is  $1.90 \times 10^{-9}$  and every solver returns an accepted convergence code.

Five random, disjoint event-half comparisons at 1.0, 2.0 and 3.0 s measure sampling reliability within each stratum. A station requires five records in each half. Both half-fields are component-aligned for storage and for constructing equal-stratum targets. The reported split-half correlation removes the weighted mean from every intersecting component pair before pooling stations, making the statistic invariant to unidentifiable component offsets. Equal-stratum sensitivity targets are formed by averaging the aligned station terms across at least two available strata, with equal weight per stratum. The primary spatial-validation procedure is then repeated for azimuth-, distance- and source-class-balanced targets; prediction errors retain station record-count weights.

## Source-conditioned station terms

Crustal, interplate and intraplate station terms use the public source labels and connected-component alignment from the path-stratified analysis. A station requires at least 20 records within a source class. Supported terms are recentered to a record-weighted zero mean for every class and period. The fixed site-parameter and site-and-regional-parameter models retain the primary variables, settings, five spatial blocks and random seed. Every source class, period, model and block is reported. Confidence intervals use 2,000 station-cluster bootstrap replicates.

Cross-validated crustal station terms are applied to the J-SHIS active-shallow response-spectrum product. Interplate and intraplate terms are applied separately to the subduction product. The two subduction calculations remain separate because the fixed-probability product does not supply the source weights needed to combine them within the hazard integral.

## Kanto 3D-simulation comparison

The publisher supplement to Sung et al. contains 250,000 Kanto grid rows for the 5.0 s adjusted site term, of which 133,151 have finite values[13]. This spatial pattern combines the simulation-derived basin adjustment and spatial nonergodic site term. The valid grid has a median nearest-neighbour spacing of 0.801 km. A station is matched to its nearest valid cell when the distance is no greater than 1.201 km, equal to 1.5 times the grid spacing. The supplement reports natural-log units; Japanese  $\log_{10}$  terms and estimates are multiplied by  $\ln(10)$ . Each pair of spatial patterns is centred by its station mean before comparison.

The primary comparison uses record-derived crustal station terms because the simulations represent crustal scenarios. Cross-validated crustal estimates, all-source record-derived terms and all-source cross-validated estimates are retained as diagnostics. Each station receives equal weight. Confidence intervals resample 31.7 km spatial blocks 2,000 times, using the Kanto correlation length reported by Sung et al. Matching, centring, source choice and interval construction were fixed before the correlations were calculated.

## Empirical station-term intervals

For each test spatial block, signed cross-validated errors from the other four blocks provide record-weighted central intervals at nominal coverage from 50% to 95%. These quantiles are added to the test-block estimate. Coverage is computed without using station terms in that block for calibration. We count intervals that exclude a unit multiplier and intervals that remain above or below a factor of 1.1. Lower and upper station terms are propagated to the matched response spectra. These intervals describe spatial-estimation error only.

Independent-model stability is calculated separately. Primary cross-validated estimates are compared at KiK-net sites with estimates from the fixed K-NET relation and disjoint earthquakes. Fold estimates are record-weighted to one value per target station and period. Pattern correlation, direction agreement, same-direction changes exceeding a factor of 1.1 and the distribution of model differences are reported. Source, ground-motion-model and J-SHIS logic-tree uncertainties lie outside both calculations.

## Reference condition conversion

J-SHIS response-spectrum maps provide spectral accelerations on engineering bedrock with  $V_S = 400 \text{ m s}^{-1}$ [14, 15]. The all-earthquake map is used for the primary result. Active-shallow and subduction maps are evaluated separately at the same cells, periods and exceedance probabilities. The station-surface factor replaces the MF2013 shallow-soil term for  $400 \text{ m s}^{-1}$  with the term evaluated at the station AVS30:

$$F_{\text{surf},i}(T) = 10^{G_s(\text{AVS30}_i,T) - G_s(400,T)}. \quad (3)$$

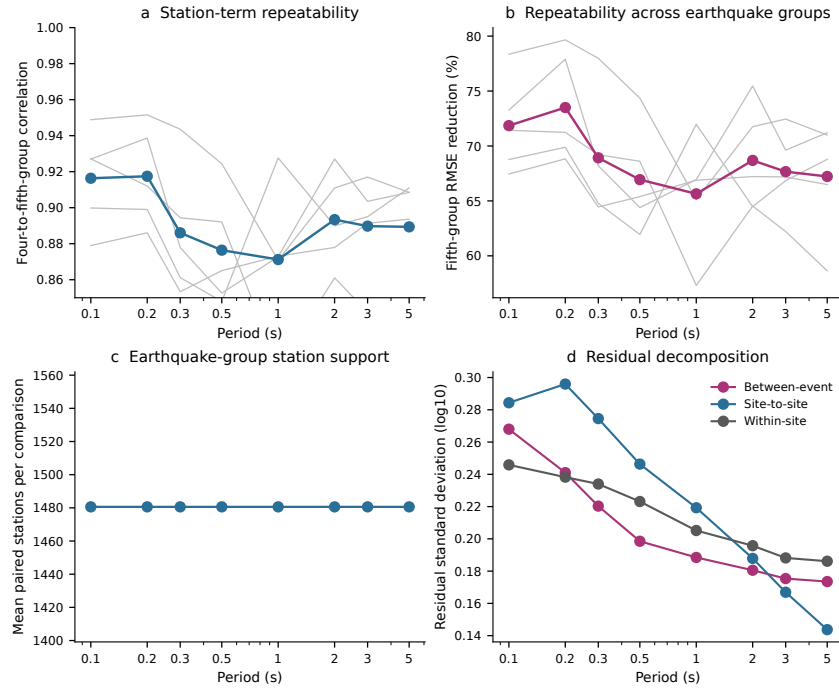
The cross-validated station factor  $10^{\widehat{\delta S}_i(T)}$  is applied after this conversion. All eight period coordinates and four 50-year exceedance probabilities are read from the J-SHIS map files. Category products retain the J-SHIS probability aggregation within each category. Their fixed-probability spectral accelerations are compared and

are not summed. The calculation is restricted to matched station cells; national-grid interpolation lies outside this analysis.

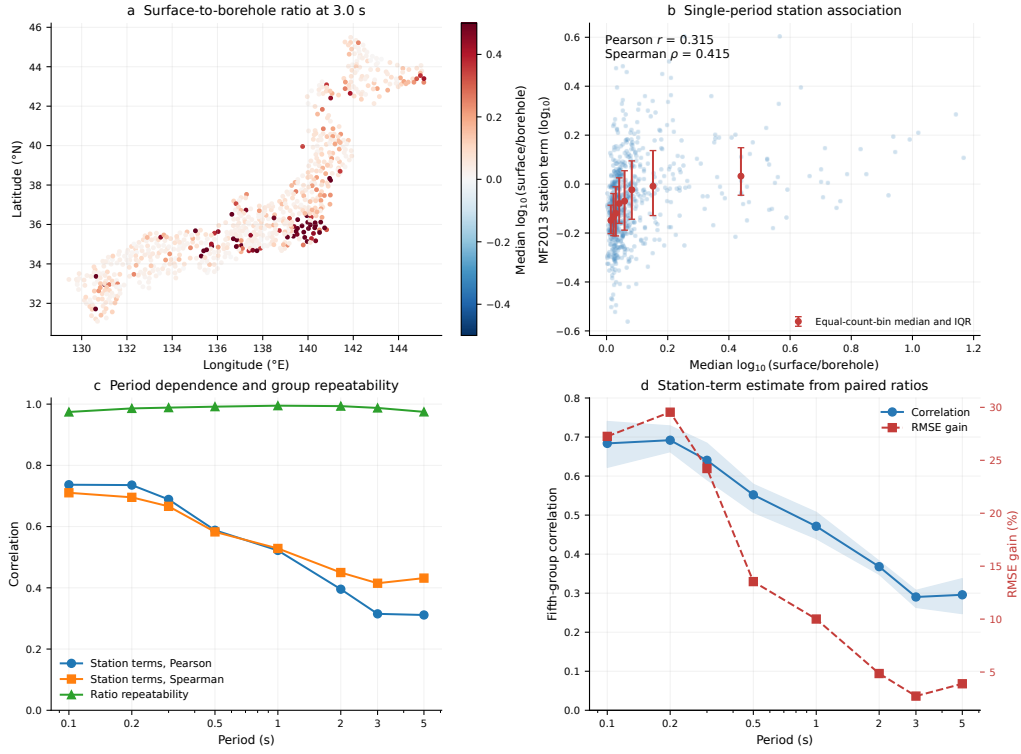
## Two-event KiK-net waveform check

The waveform check uses 460 paired surface and downhole records from two KiK-net earthquakes[16]. The first set contains 288 converted miniSEED pairs and the second contains 172 original NIED ASCII pairs converted with the scale factors in their headers. Each trace is demeaned and detrended. Horizontal peak amplitude is calculated from the two horizontal components. Surface-to-downhole Fourier amplitude ratios are evaluated from 0.2 to 20 Hz. This check is independent of the station residual model.

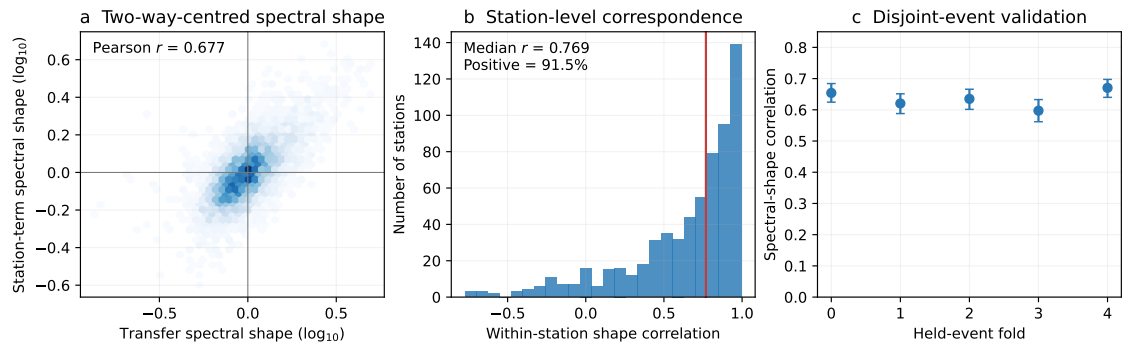
## Supplementary Figures



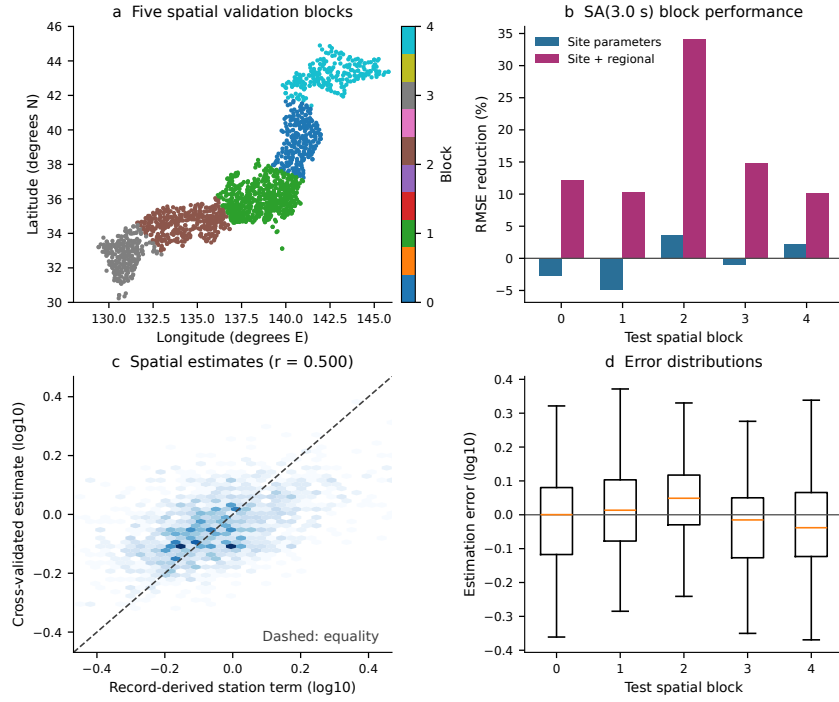
**Supplementary Figure 1. Station-term repeatability across independent earthquake groups.** **a**, correlations between station terms estimated from four earthquake groups and the fifth group; grey lines show individual comparisons and the blue line shows their mean. **b**, root mean squared error reduction for fifth-group station terms relative to a zero station term. **c**, mean number of stations meeting the two record thresholds. **d**, between-event, site-to-site and within-site residual amplitudes from the primary MF2013 decomposition.



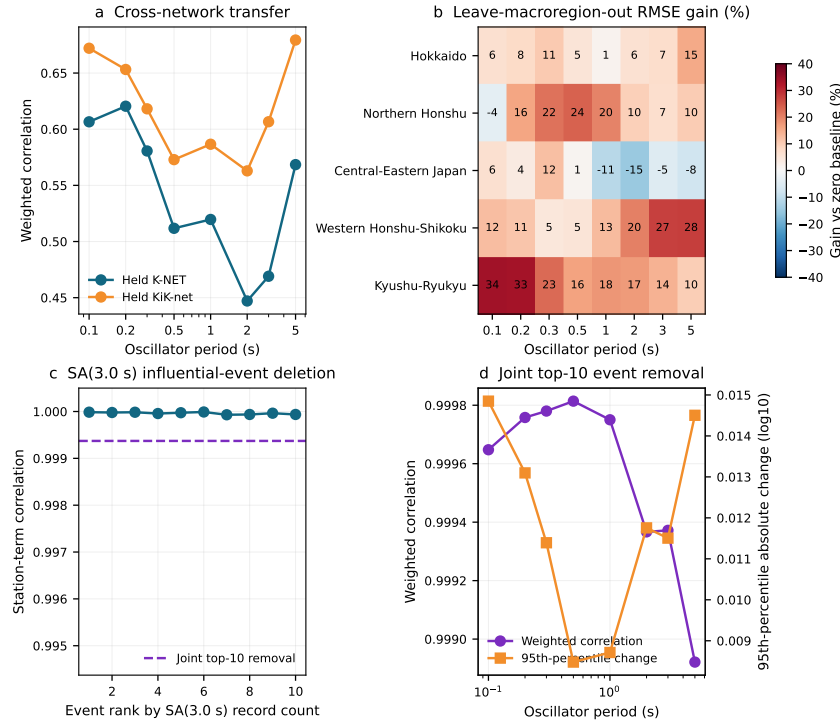
**Supplementary Figure 2. National KiK-net paired-sensor validation.** **a**, median surface-to-borehole RotD100 response-spectrum ratio at 3.0 s for 699 sites. **b**, surface station terms against paired response-spectrum ratios at supported sites; red points and bars show equal-count-bin medians and interquartile ranges, without a fitted line. **c**, single-period correspondence and ratio repeatability across periods. **d**, fifth-group station-term estimates from response-spectrum ratios calculated with four independent earthquake groups.



**Supplementary Figure 3. Paired-sensor spectral-shape validation.** **a**, two-way-centred station-term and surface-to-borehole response-spectrum-ratio shapes across eight periods. **b**, distribution of within-station shape correlations. **c**, correlations when ratio spectra and station-term spectra are estimated from disjoint earthquake groups; bars are station-cluster bootstrap 95% intervals.

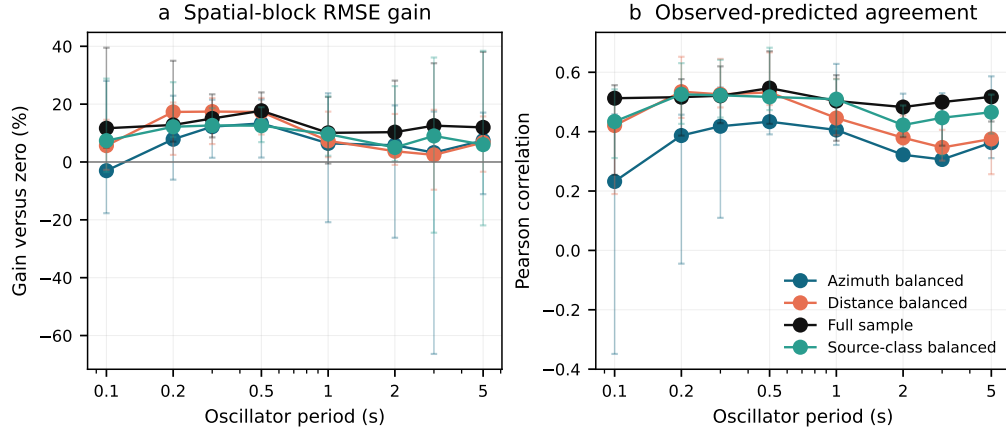


**Supplementary Figure 4. Spatial estimation of SA(3.0 s) station terms.** **a**, five station blocks used for cross-validation. **b**, root mean squared error reduction in each test block for the site-parameter and site-and-regional-parameter models. **c**, record-derived station terms and cross-validated site-and-regional-parameter estimates; the dashed diagonal marks equality and is not a fitted relation. **d**, estimation-error distributions by spatial block; boxes show interquartile ranges and whiskers extend to 1.5 times the interquartile range.

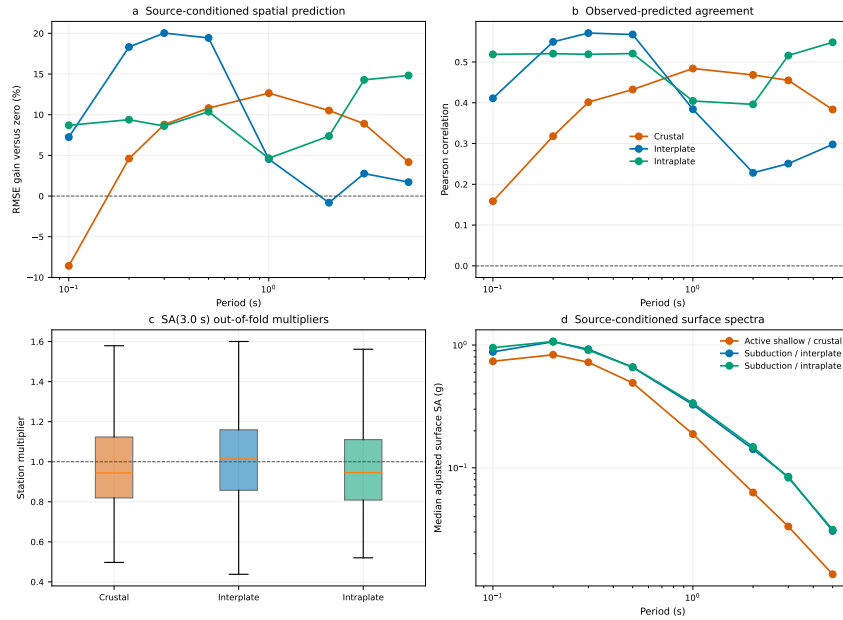


**Supplementary Figure 5. Network, geographic and event-influence tests.** **a**, cross-network comparison using the common variable set. **b**, root mean squared error gain when each fixed macroregion provides the test stations. **c**, SA(3.0 s) station-term correlation after deleting each of the ten highest-count earthquakes; the dashed line shows their joint deletion. **d**, station-term correlation and weighted 95th-percentile absolute change after jointly deleting the ten earthquakes at every period.

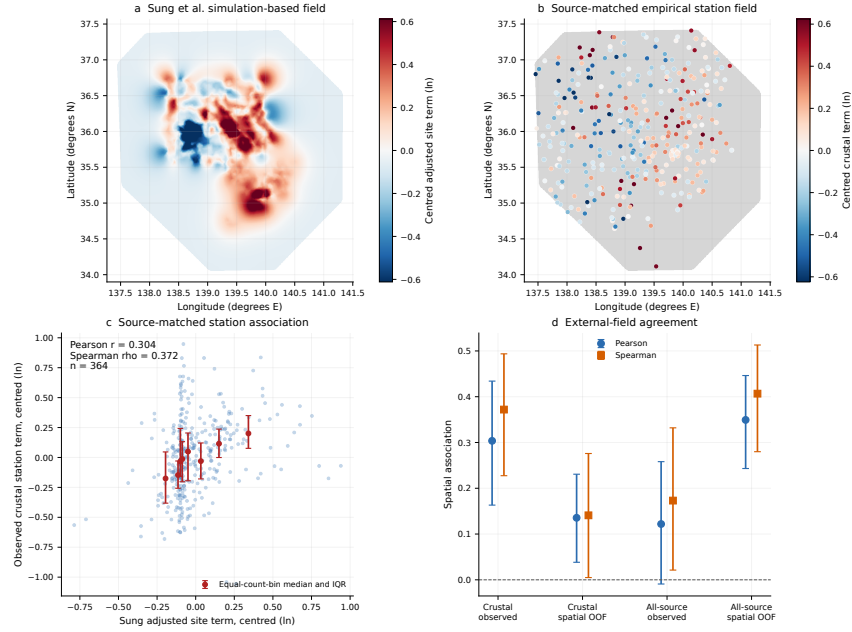




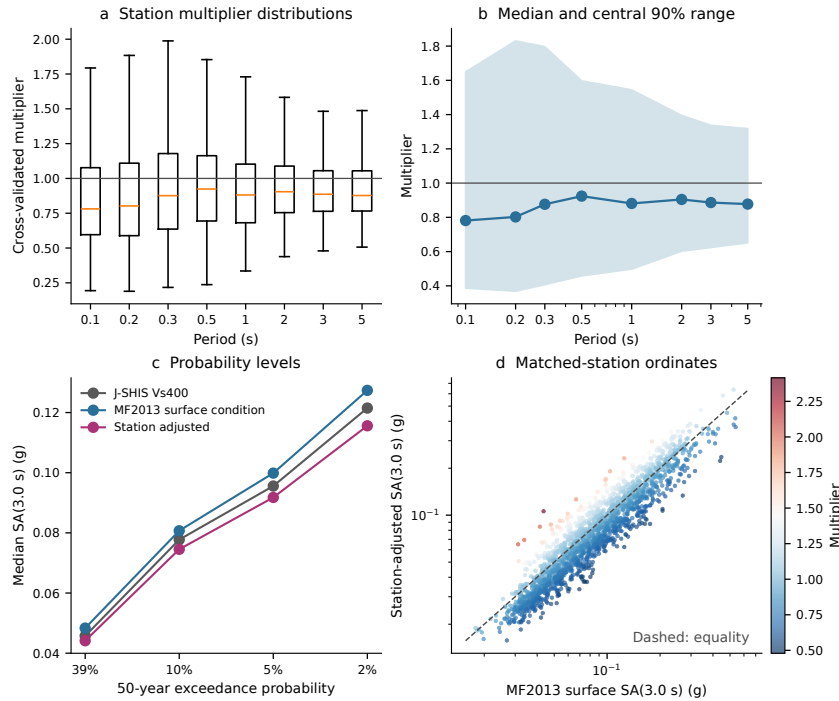
**Supplementary Figure 6. Spatial estimation of equal-stratum station terms.** **a**, overall spatial-block root mean squared error gain relative to zero. **b**, record-derived versus estimated correlation. Vertical bars span the five test-block results; circles show pooled results. The full-sample terms are included for reference.



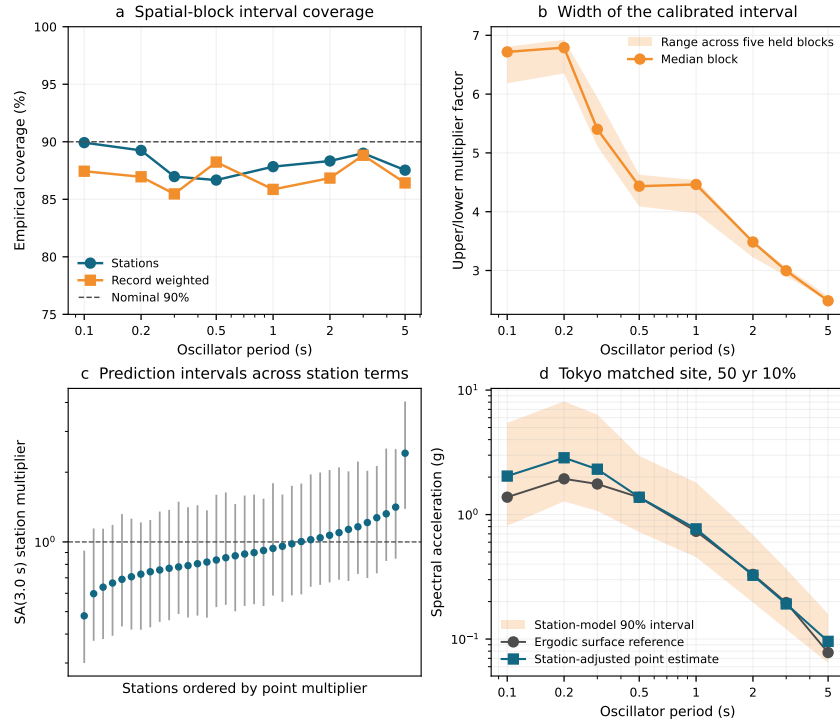
**Supplementary Figure 7. Source-conditioned station-term estimation and spectrum sensitivity.** **a**, spatial-block root mean squared error gain for crustal, interplate and intraplate station terms. **b**, record-derived versus estimated correlations. **c**, SA(3.0 s) cross-validated station multipliers; boxes show interquartile ranges and whiskers extend to 1.5 times the interquartile range. **d**, matched-station median adjusted spectra after applying each source-conditioned estimate to the corresponding J-SHIS product. Interplate and intraplate sensitivities are evaluated separately.



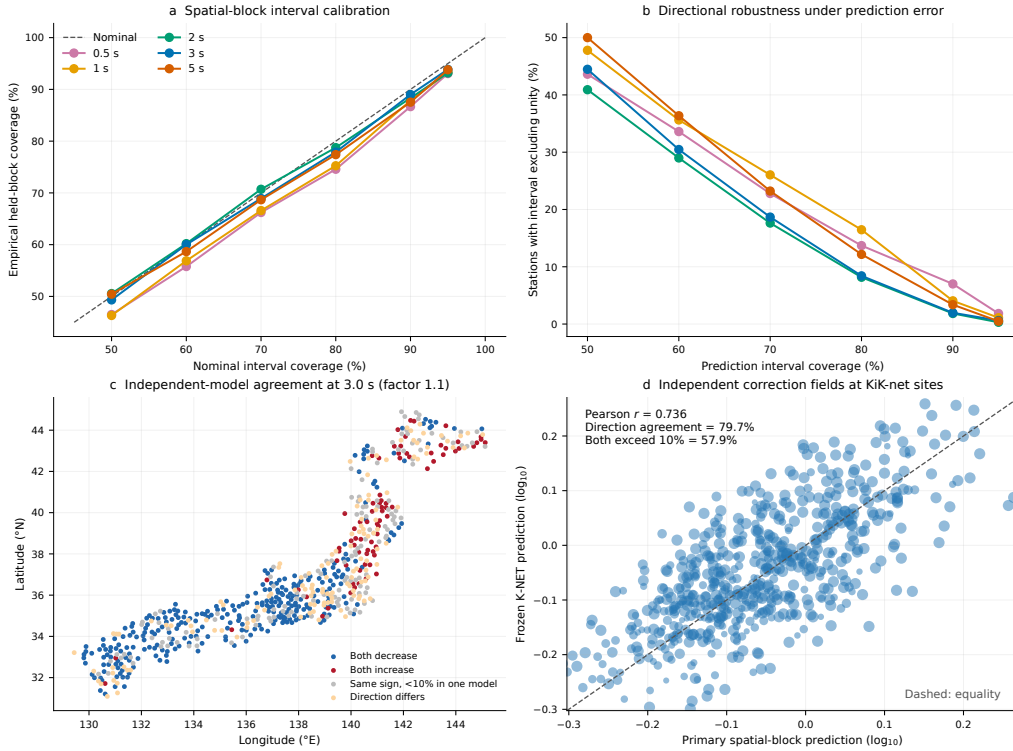
**Supplementary Figure 8. Independent Kanto 3D-simulation comparison at 5.0 s.** **a**, centred publisher-supplied adjusted site-term pattern from Sung et al. **b**, source-matched crustal station terms at cells passing the fixed nearest-grid rule. **c**, centred simulation and record-derived terms at 364 stations; red points and bars show equal-count-bin medians and interquartile ranges, without a fitted line. **d**, Pearson and Spearman correlations for record-derived and cross-validated station terms; bars are 95% intervals from 2,000 resamples of 31.7 km spatial blocks.



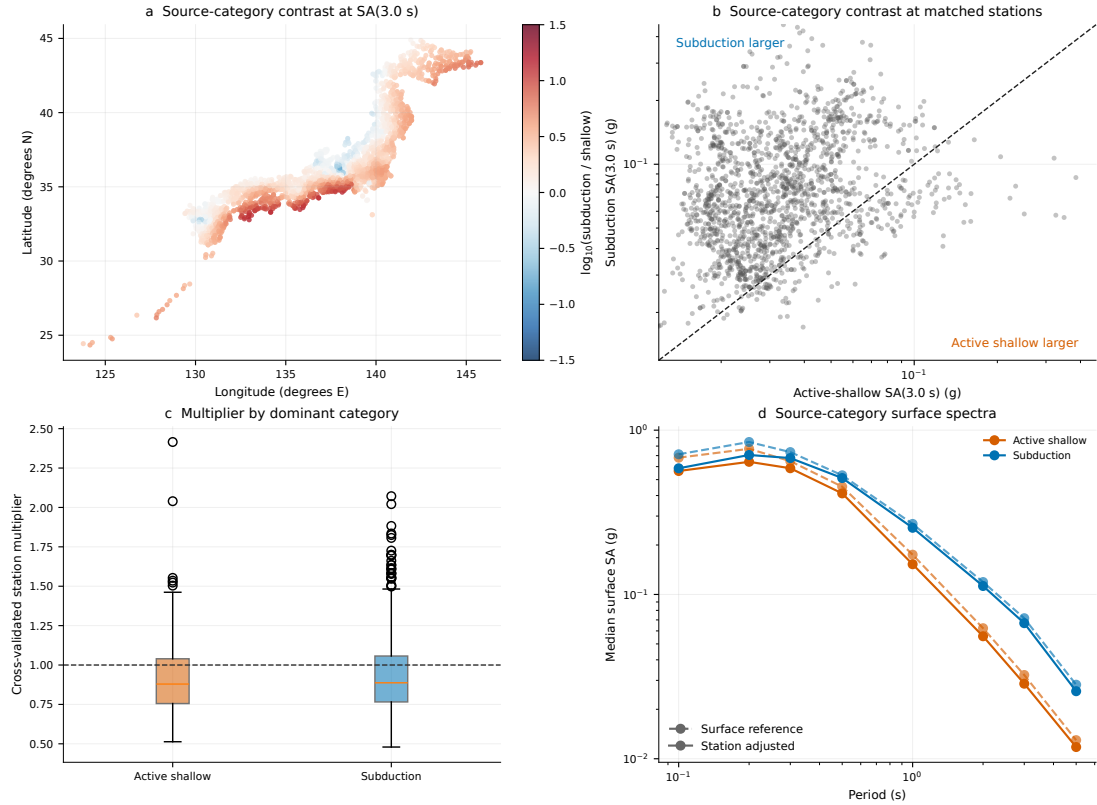
**Supplementary Figure 9. Matched-station response-spectrum sensitivity.** **a**, cross-validated station multiplier distributions at the eight direct periods. **b**, median and 5th–95th percentile multiplier range. **c**, median  $SA(3.0\text{ s})$  at four 50-year exceedance probabilities for the J-SHIS  $V_S = 400\text{ m s}^{-1}$  reference, the MF2013 surface condition and the station-adjusted surface condition. **d**, paired MF2013 surface and station-adjusted  $SA(3.0\text{ s})$  values at the 50-year 10% level; colour denotes the station multiplier and the dashed diagonal marks equality.



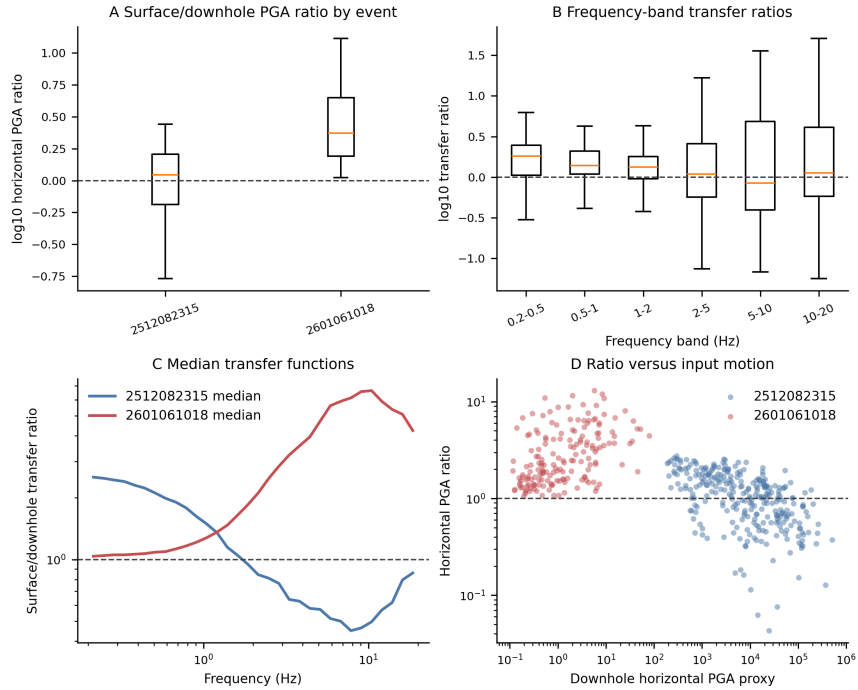
**Supplementary Figure 10. Spatial-estimation uncertainty.** **a**, empirical coverage of the nominal 90% interval by period. **b**, median interval factor span and its range across the five test blocks. **c**, point multipliers and empirical intervals for 35 ordered SA(3.0 s) stations. **d**, surface-condition and station-adjusted spectra with the estimation interval at the Tokyo-nearest matched station. The interval excludes source, ground-motion-model and J-SHIS logic-tree uncertainty.



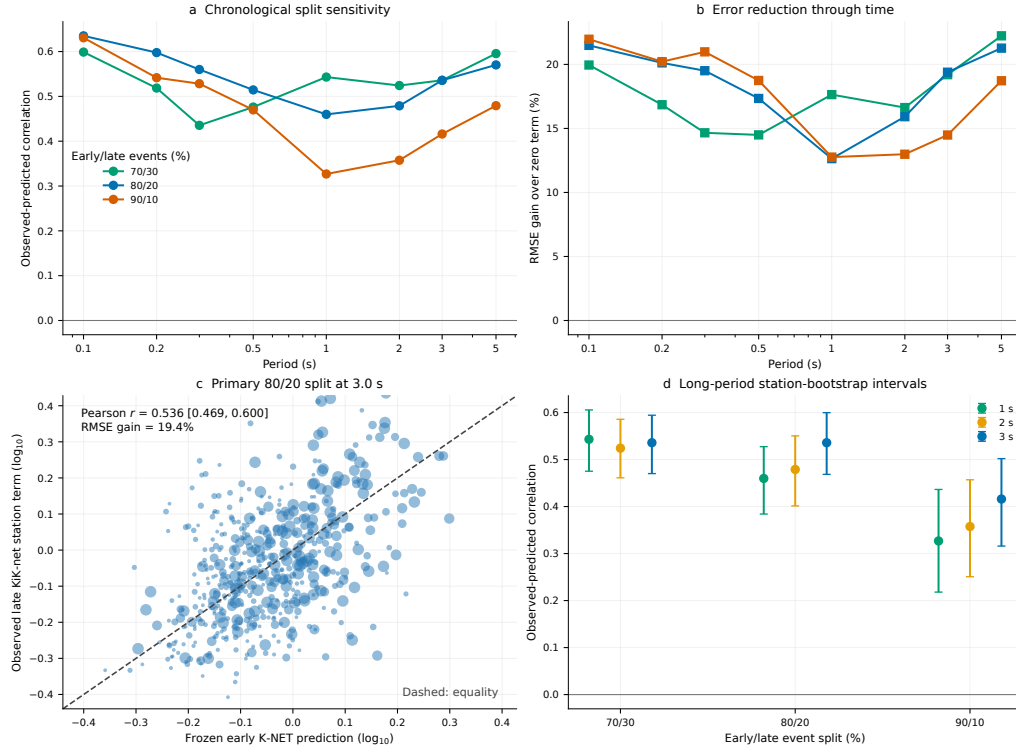
**Supplementary Figure 11. Uncertainty and independent-model stability of station adjustments.** **a**, empirical spatial-block coverage against nominal coverage. **b**, fraction of station intervals excluding a unit multiplier. **c**, 3.0 s KiK-net sites classified by agreement between the primary and fixed K-NET station-term estimates. **d**, the two independent sets of estimates at 3.0 s; the dashed diagonal marks equality. Estimation intervals and model agreement quantify different uncertainty components.



**Supplementary Figure 12. J-SHIS source-category response-spectrum sensitivity.** **a**, spatial contrast between subduction and active-shallow SA(3.0 s) at the 50-year 10% level. **b**, paired source-category spectral accelerations; the dashed diagonal marks equality. **c**, fixed station multipliers grouped by the category with the larger J-SHIS value. **d**, matched-station median surface spectra for the two categories; dashed lines show the MF2013 surface condition and solid lines include the station adjustment. Category values are evaluated separately and are not summed.



**Supplementary Figure 13. KiK-net surface and downhole waveform comparison.** **a**, horizontal surface-to-downhole peak acceleration ratios by earthquake. **b**, frequency-band Fourier amplitude ratios. **c**, median transfer functions for the two event sets. **d**, horizontal peak acceleration ratio against the downhole input-motion proxy. Ratios are shown in  $\log_{10}$  units in panels **a** and **b**. Dashed horizontal lines denote equal surface and downhole amplitudes.



**Supplementary Figure 14. Chronological cross-network station-term test.** **a**, K-NET-to-KiK-net correlations for fixed early/late earthquake splits. **b**, root mean squared error reduction relative to a zero station term. **c**, fixed estimates from the earliest 80% of K-NET earthquakes against KiK-net terms estimated from the latest 20% at 3.0 s; the dashed diagonal marks equality. **d**, station-bootstrap 95% intervals at 1–3 s. The reverse direction is reported in Supplementary Table 20.

## Supplementary Tables

**Supplementary Table 1. Primary residual decomposition**

| Period<br>(s) | Records | Global offset | Between-event SD<br>log <sub>10</sub> units | Site-to-site SD | Within-site SD |
|---------------|---------|---------------|---------------------------------------------|-----------------|----------------|
| 0.1           | 222,664 | 0.284         | 0.268                                       | 0.284           | 0.246          |
| 0.2           | 222,664 | 0.215         | 0.241                                       | 0.296           | 0.238          |
| 0.3           | 222,664 | 0.172         | 0.220                                       | 0.275           | 0.234          |
| 0.5           | 222,664 | 0.106         | 0.199                                       | 0.246           | 0.223          |
| 1.0           | 222,664 | 0.017         | 0.188                                       | 0.219           | 0.205          |
| 2.0           | 222,664 | -0.039        | 0.181                                       | 0.188           | 0.196          |
| 3.0           | 222,664 | -0.085        | 0.175                                       | 0.167           | 0.188          |
| 5.0           | 222,664 | -0.134        | 0.174                                       | 0.144           | 0.186          |

**Supplementary Table 2.** Station-term repeatability across independent earthquake groups

| Period<br>(s) | Four-to-fifth-group<br>correlation | Fifth-group RMSE<br>reduction (%) | Mean paired<br>stations | Mean fifth-group<br>earthquakes |
|---------------|------------------------------------|-----------------------------------|-------------------------|---------------------------------|
| 0.1           | 0.916                              | 71.9                              | 1,481                   | 347.4                           |
| 0.2           | 0.917                              | 73.5                              | 1,481                   | 347.4                           |
| 0.3           | 0.886                              | 68.9                              | 1,481                   | 347.4                           |
| 0.5           | 0.876                              | 66.9                              | 1,481                   | 347.4                           |
| 1.0           | 0.871                              | 65.6                              | 1,481                   | 347.4                           |
| 2.0           | 0.893                              | 68.7                              | 1,481                   | 347.4                           |
| 3.0           | 0.890                              | 67.7                              | 1,481                   | 347.4                           |
| 5.0           | 0.889                              | 67.2                              | 1,481                   | 347.4                           |

**Supplementary Table 3.** Zhao 2006 ground-motion-model sensitivity

| Period<br>(s) | MF2013–Zhao<br>correlation | Median absolute<br>difference ( $\log_{10}$ ) | Zhao event<br>correlation | Zhao spatial<br>RMSE gain (%) |
|---------------|----------------------------|-----------------------------------------------|---------------------------|-------------------------------|
| 0.1           | 0.953                      | 0.064                                         | 0.940                     | 16.9                          |
| 0.2           | 0.921                      | 0.093                                         | 0.941                     | 20.8                          |
| 0.3           | 0.919                      | 0.081                                         | 0.913                     | 18.9                          |
| 0.5           | 0.923                      | 0.073                                         | 0.909                     | 20.8                          |
| 1.0           | 0.889                      | 0.094                                         | 0.915                     | 22.9                          |
| 2.0           | 0.801                      | 0.144                                         | 0.948                     | 32.1                          |
| 3.0           | 0.770                      | 0.151                                         | 0.950                     | 37.7                          |
| 5.0           | 0.697                      | 0.125                                         | 0.946                     | 39.2                          |

**Supplementary Table 4.** KiK-net paired-sensor spectral validation

| Period<br>(s) | Station term versus paired ratio |          |                        | Ratio group<br>repeatability | Fifth-group term<br>correlation | Fifth-group RMSE<br>gain (%) |
|---------------|----------------------------------|----------|------------------------|------------------------------|---------------------------------|------------------------------|
|               | Pearson                          | Spearman | Depth-adjusted Pearson |                              |                                 |                              |
| 0.1           | 0.737                            | 0.710    | 0.739                  | 0.974                        | 0.684                           | 27.3                         |
| 0.2           | 0.736                            | 0.696    | 0.737                  | 0.986                        | 0.692                           | 29.5                         |
| 0.3           | 0.689                            | 0.666    | 0.703                  | 0.989                        | 0.640                           | 24.2                         |
| 0.5           | 0.588                            | 0.583    | 0.630                  | 0.992                        | 0.552                           | 13.5                         |
| 1.0           | 0.522                            | 0.529    | 0.585                  | 0.995                        | 0.471                           | 10.0                         |
| 2.0           | 0.395                            | 0.450    | 0.437                  | 0.993                        | 0.368                           | 4.9                          |
| 3.0           | 0.315                            | 0.415    | 0.327                  | 0.988                        | 0.290                           | 2.7                          |
| 5.0           | 0.311                            | 0.432    | 0.342                  | 0.975                        | 0.296                           | 3.9                          |

**Supplementary Table 5.** Cross-validated spatial estimation of station terms

| Period<br>(s) | Site parameters | Site and regional parameters | Record-derived versus estimated<br>correlation |
|---------------|-----------------|------------------------------|------------------------------------------------|
|               | RMSE gain (%)   | RMSE gain (%)                |                                                |
| 0.1           | -5.2            | 11.7                         | 0.513                                          |
| 0.2           | 4.3             | 12.8                         | 0.516                                          |
| 0.3           | 10.3            | 15.0                         | 0.521                                          |
| 0.5           | 9.0             | 17.7                         | 0.547                                          |
| 1.0           | 4.6             | 10.1                         | 0.503                                          |
| 2.0           | 0.0             | 10.3                         | 0.483                                          |
| 3.0           | -2.3            | 12.6                         | 0.500                                          |
| 5.0           | -3.5            | 12.0                         | 0.517                                          |

**Supplementary Table 6.** SA(3.0 s) network and fixed-macroregion comparison

| Test group             | Full variable set<br>RMSE gain (%) | Common variable set<br>RMSE gain (%) | Common variable set<br>correlation |
|------------------------|------------------------------------|--------------------------------------|------------------------------------|
| K-NET                  | 4.1                                | 8.7                                  | 0.469                              |
| KiK-net                | 21.2                               | 20.5                                 | 0.607                              |
| Hokkaido               | 7.2                                | 6.9                                  | 0.284                              |
| Northern Honshu        | 5.4                                | 6.8                                  | 0.304                              |
| Central-Eastern Japan  | 6.2                                | -5.2                                 | 0.402                              |
| Western Honshu–Shikoku | 28.9                               | 27.5                                 | 0.386                              |
| Kyushu–Ryukyu          | 13.4                               | 13.8                                 | 0.372                              |

**Supplementary Table 7.** Joint deletion of the ten highest-count events

| Period<br>(s) | Station-term<br>correlation | Weighted 95th-percentile<br>absolute change ( $\log_{10}$ ) |
|---------------|-----------------------------|-------------------------------------------------------------|
| 0.1           | 0.99965                     | 0.0148                                                      |
| 0.2           | 0.99976                     | 0.0131                                                      |
| 0.3           | 0.99978                     | 0.0114                                                      |
| 0.5           | 0.99981                     | 0.0085                                                      |
| 1.0           | 0.99975                     | 0.0087                                                      |
| 2.0           | 0.99937                     | 0.0118                                                      |
| 3.0           | 0.99937                     | 0.0115                                                      |
| 5.0           | 0.99892                     | 0.0145                                                      |

**Supplementary Table 8.** SA(3.0 s) path and source stratification

| Stratification      | Stratum    | Paired stations | All-record<br>correlation | Within-component<br>split-half correlation | 95th percentile<br>difference |
|---------------------|------------|-----------------|---------------------------|--------------------------------------------|-------------------------------|
| Hypocentral azimuth | 000–090°   | 938             | 0.280                     | 0.884                                      | 0.474                         |
| Hypocentral azimuth | 090–180°   | 448             | 0.678                     | 0.764                                      | 0.236                         |
| Hypocentral azimuth | 180–270°   | 1,089           | 0.777                     | 0.969                                      | 0.317                         |
| Hypocentral azimuth | 270–360°   | 1,345           | 0.886                     | 0.945                                      | 0.168                         |
| Fault distance      | 0–50 km    | 138             | 0.644                     | 0.631                                      | 0.289                         |
| Fault distance      | 50–100 km  | 680             | 0.496                     | 0.793                                      | 0.427                         |
| Fault distance      | 100–200 km | 1,515           | 0.974                     | 0.951                                      | 0.080                         |
| Fault distance      | 200–300 km | 1,213           | 0.963                     | 0.968                                      | 0.092                         |
| Source type         | Crustal    | 1,494           | 0.942                     | 0.905                                      | 0.120                         |
| Source type         | Interplate | 960             | 0.934                     | 0.958                                      | 0.073                         |
| Source type         | Intraplate | 1,379           | 0.924                     | 0.902                                      | 0.131                         |

**Supplementary Table 9.** Equal-stratum spatial-estimation sensitivity

| Target terms      | Period<br>(s) | Stations | Overall RMSE<br>reduction (%) | Overall<br>correlation | Minimum block<br>reduction (%) |
|-------------------|---------------|----------|-------------------------------|------------------------|--------------------------------|
| Azimuth balanced  | 1.0           | 1,220    | 6.4                           | 0.406                  | -20.8                          |
| Azimuth balanced  | 2.0           | 1,220    | 5.8                           | 0.322                  | -26.2                          |
| Azimuth balanced  | 3.0           | 1,220    | 3.2                           | 0.306                  | -66.4                          |
| Distance balanced | 1.0           | 1,285    | 7.3                           | 0.446                  | 1.7                            |
| Distance balanced | 2.0           | 1,285    | 3.7                           | 0.379                  | -1.0                           |
| Distance balanced | 3.0           | 1,285    | 2.5                           | 0.347                  | -9.6                           |
| Source balanced   | 1.0           | 1,369    | 9.6                           | 0.509                  | 2.2                            |
| Source balanced   | 2.0           | 1,369    | 5.0                           | 0.422                  | 0.3                            |
| Source balanced   | 3.0           | 1,369    | 9.0                           | 0.447                  | -24.5                          |

**Supplementary Table 10.** Primary full-sample matched-station 50-year 10 percent response spectra

| Period<br>(s) | Station multiplier |        |       | Official $V_S400$<br>median spectral acceleration | MF2013 surface<br>spectral acceleration (g) | Station adjusted<br>acceleration (g) |
|---------------|--------------------|--------|-------|---------------------------------------------------|---------------------------------------------|--------------------------------------|
|               | 5th                | Median | 95th  |                                                   |                                             |                                      |
| 0.1           | 0.386              | 0.781  | 1.650 | 0.946                                             | 0.978                                       | 0.817                                |
| 0.2           | 0.368              | 0.802  | 1.830 | 1.068                                             | 1.154                                       | 0.936                                |
| 0.3           | 0.407              | 0.876  | 1.797 | 0.914                                             | 1.022                                       | 0.891                                |
| 0.5           | 0.458              | 0.924  | 1.597 | 0.652                                             | 0.737                                       | 0.668                                |
| 1.0           | 0.498              | 0.881  | 1.544 | 0.315                                             | 0.341                                       | 0.308                                |
| 2.0           | 0.602              | 0.904  | 1.395 | 0.132                                             | 0.140                                       | 0.130                                |
| 3.0           | 0.624              | 0.886  | 1.338 | 0.078                                             | 0.081                                       | 0.075                                |
| 5.0           | 0.651              | 0.877  | 1.318 | 0.030                                             | 0.031                                       | 0.028                                |

**Supplementary Table 11.** Empirical station-model prediction intervals

| Period<br>(s) | Station<br>coverage (%) | Record-weighted<br>coverage (%) | Median upper/lower<br>multiplier factor |
|---------------|-------------------------|---------------------------------|-----------------------------------------|
| 0.1           | 89.9                    | 87.4                            | 6.72                                    |
| 0.2           | 89.3                    | 87.0                            | 6.79                                    |
| 0.3           | 87.0                    | 85.5                            | 5.40                                    |
| 0.5           | 86.7                    | 88.2                            | 4.43                                    |
| 1.0           | 87.8                    | 85.9                            | 4.32                                    |
| 2.0           | 88.3                    | 86.8                            | 3.49                                    |
| 3.0           | 89.0                    | 88.8                            | 3.00                                    |
| 5.0           | 87.5                    | 86.4                            | 2.48                                    |

**Supplementary Table 12.** City-nearest SA(3.0 s) station cases

| City    | Station | Distance<br>(km) | Multiplier | Change<br>(%) | Surface SA<br>(g) | Adjusted SA<br>(g) |
|---------|---------|------------------|------------|---------------|-------------------|--------------------|
| Tokyo   | TKY007  | 2.4              | 0.975      | -2.5          | 0.196             | 0.192              |
| Osaka   | OSK005  | 3.6              | 0.885      | -11.5         | 0.223             | 0.197              |
| Sapporo | HKD180  | 8.6              | 1.312      | +31.2         | 0.124             | 0.163              |
| Sendai  | MYG013  | 5.0              | 0.890      | -11.0         | 0.131             | 0.116              |
| Fukuoka | FKOH03  | 14.1             | 0.748      | -25.2         | 0.035             | 0.026              |

**Supplementary Table 13.** Sequential record selection

| Stage                                                | Records | Excluded | Earthquakes | Stations |
|------------------------------------------------------|---------|----------|-------------|----------|
| Public <b>sub1-v2024</b> archive                     | 333,808 | 0        | 1,840       | 2,581    |
| Ground-surface installation                          | 231,380 | 102,428  | 1,840       | 1,882    |
| Finite F-net $M_w$                                   | 222,705 | 8,675    | 1,737       | 1,881    |
| Supported source class and positive distance         | 222,705 | 0        | 1,737       | 1,881    |
| Positive AVS30                                       | 222,664 | 41       | 1,737       | 1,880    |
| Finite RotD100 and MF2013 predictions at all periods | 222,664 | 0        | 1,737       | 1,880    |



**Supplementary Table 14.** Response-component sensitivity

| Period<br>(s) | Station-field<br>correlation | Weighted difference<br>RMSE | 95th percentile<br>absolute difference | Median record<br>RotD50/RotD100 |
|---------------|------------------------------|-----------------------------|----------------------------------------|---------------------------------|
| 0.1           | 0.9995                       | 0.0076                      | 0.0179                                 | 0.853                           |
| 0.2           | 0.9995                       | 0.0081                      | 0.0183                                 | 0.839                           |
| 0.3           | 0.9994                       | 0.0079                      | 0.0176                                 | 0.834                           |
| 0.5           | 0.9993                       | 0.0073                      | 0.0177                                 | 0.831                           |
| 1.0           | 0.9990                       | 0.0077                      | 0.0197                                 | 0.826                           |
| 2.0           | 0.9988                       | 0.0074                      | 0.0194                                 | 0.820                           |
| 3.0           | 0.9986                       | 0.0071                      | 0.0180                                 | 0.819                           |
| 5.0           | 0.9983                       | 0.0064                      | 0.0177                                 | 0.820                           |

**Supplementary Table 15.** SA(3.0 s) spatial-model complexity sensitivity

| Setting                  | Changed parameter             | RMSE gain (%) | Correlation |
|--------------------------|-------------------------------|---------------|-------------|
| Primary                  | None                          | 12.56         | 0.500       |
| Fewer leaves             | Maximum leaf nodes = 7        | 13.98         | 0.512       |
| More leaves              | Maximum leaf nodes = 31       | 10.45         | 0.467       |
| Larger minimum leaf      | Minimum samples per leaf = 70 | 10.71         | 0.483       |
| Stronger regularisation  | L2 regularisation = 1.0       | 12.29         | 0.496       |
| Shorter iteration budget | Maximum iterations = 200      | 12.85         | 0.497       |

**Supplementary Table 16.** MF2013 regression-domain sensitivity including the Kanno-PGA cutoff

| Period<br>(s) | Station-field<br>correlation | 95th-percentile<br>difference ( $\log_{10}$ ) | Event-group<br>correlation | Spatial RMSE<br>gain (%) | OOF-prediction<br>correlation | Restricted multiplier<br>5th–95th percentile |
|---------------|------------------------------|-----------------------------------------------|----------------------------|--------------------------|-------------------------------|----------------------------------------------|
| 0.1           | 0.963                        | 0.144                                         | 0.916                      | -2.3                     | 0.671                         | 0.485–1.871                                  |
| 0.2           | 0.973                        | 0.128                                         | 0.909                      | 12.2                     | 0.766                         | 0.436–2.018                                  |
| 0.3           | 0.969                        | 0.117                                         | 0.903                      | 16.3                     | 0.720                         | 0.454–1.995                                  |
| 0.5           | 0.965                        | 0.117                                         | 0.884                      | 18.5                     | 0.756                         | 0.481–1.785                                  |
| 1.0           | 0.959                        | 0.109                                         | 0.829                      | 9.3                      | 0.666                         | 0.538–1.679                                  |
| 2.0           | 0.956                        | 0.094                                         | 0.777                      | 5.8                      | 0.615                         | 0.596–1.541                                  |
| 3.0           | 0.959                        | 0.082                                         | 0.762                      | 4.1                      | 0.648                         | 0.660–1.503                                  |
| 5.0           | 0.924                        | 0.098                                         | 0.719                      | 7.6                      | 0.637                         | 0.707–1.428                                  |

**Supplementary Table 17.** KiK-net station-term and surface–borehole spectral-shape correspondence

| Event scope | Stations | Shape<br>correlation | Cluster-bootstrap<br>95% interval | Median station<br>correlation | Positive stations<br>(%) |
|-------------|----------|----------------------|-----------------------------------|-------------------------------|--------------------------|
| All events  | 639      | 0.677                | 0.648–0.704                       | 0.769                         | 91.5                     |
| Held fold 1 | 603      | 0.654                | 0.625–0.684                       | 0.732                         | 91.5                     |
| Held fold 2 | 568      | 0.620                | 0.588–0.651                       | 0.701                         | 88.6                     |
| Held fold 3 | 607      | 0.635                | 0.601–0.666                       | 0.734                         | 89.3                     |
| Held fold 4 | 537      | 0.597                | 0.562–0.633                       | 0.666                         | 88.1                     |
| Held fold 5 | 605      | 0.670                | 0.640–0.697                       | 0.751                         | 91.2                     |
| Fold mean   | 584      | 0.635                | —                                 | 0.717                         | 89.7                     |

**Supplementary Table 18.** Cross-network and independent-earthquake station-term estimation

| Period<br>(s) | K-NET to KiK-net, site parameters |               | K-NET to KiK-net, site + regional |               | KiK-net to K-NET, site + regional |               |
|---------------|-----------------------------------|---------------|-----------------------------------|---------------|-----------------------------------|---------------|
|               | Correlation                       | RMSE gain (%) | Correlation                       | RMSE gain (%) | Correlation                       | RMSE gain (%) |
| 0.1           | 0.368                             | 6.2           | 0.641                             | 22.3          | 0.549                             | 13.1          |
| 0.2           | 0.394                             | 6.2           | 0.580                             | 20.0          | 0.490                             | 16.0          |
| 0.3           | 0.439                             | 12.2          | 0.570                             | 22.0          | 0.390                             | 7.9           |
| 0.5           | 0.416                             | 9.6           | 0.543                             | 19.8          | 0.278                             | -5.0          |
| 1.0           | 0.416                             | 9.5           | 0.575                             | 20.3          | 0.360                             | -0.3          |
| 2.0           | 0.417                             | 11.5          | 0.567                             | 18.5          | 0.430                             | 2.3           |
| 3.0           | 0.439                             | 13.0          | 0.598                             | 20.8          | 0.462                             | 4.0           |
| 5.0           | 0.447                             | 10.3          | 0.642                             | 25.9          | 0.539                             | 14.3          |

**Supplementary Table 19.** Agreement between primary and fixed K-NET station-term estimates at KiK-net sites

| Period<br>(s) | Pattern<br>correlation | Direction<br>agreement (%) | Same-direction change<br>above 10% (%) | Median absolute<br>difference ( $\log_{10}$ ) |
|---------------|------------------------|----------------------------|----------------------------------------|-----------------------------------------------|
| 0.1           | 0.709                  | 81.1                       | 68.5                                   | 0.100                                         |
| 0.2           | 0.692                  | 80.3                       | 67.3                                   | 0.122                                         |
| 0.3           | 0.680                  | 76.8                       | 63.4                                   | 0.123                                         |
| 0.5           | 0.690                  | 76.8                       | 63.1                                   | 0.097                                         |
| 1.0           | 0.698                  | 76.4                       | 57.0                                   | 0.079                                         |
| 2.0           | 0.713                  | 77.2                       | 56.2                                   | 0.061                                         |
| 3.0           | 0.736                  | 79.7                       | 57.9                                   | 0.053                                         |
| 5.0           | 0.752                  | 80.4                       | 59.6                                   | 0.046                                         |

**Supplementary Table 20.** Chronological K-NET-to-KiK-net station-term test

| Early/late events<br>(%) | Period<br>(s) | Target stations | Correlation | Correlation 95% interval | RMSE gain<br>(%) | RMSE-gain 95% interval<br>(%) |
|--------------------------|---------------|-----------------|-------------|--------------------------|------------------|-------------------------------|
| 70/30                    | 1.0           | 593             | 0.543       | 0.475–0.606              | 17.6             | 10.1–24.4                     |
| 70/30                    | 2.0           | 593             | 0.524       | 0.461–0.586              | 16.6             | 9.7–23.4                      |
| 70/30                    | 3.0           | 593             | 0.536       | 0.470–0.594              | 19.2             | 13.6–24.2                     |
| 80/20                    | 0.1           | 547             | 0.635       | 0.587–0.679              | 21.5             | 16.1–26.6                     |
| 80/20                    | 0.2           | 547             | 0.598       | 0.537–0.652              | 20.1             | 14.4–25.4                     |
| 80/20                    | 0.3           | 547             | 0.560       | 0.495–0.620              | 19.5             | 12.4–26.7                     |
| 80/20                    | 0.5           | 547             | 0.514       | 0.448–0.577              | 17.3             | 10.1–23.9                     |
| 80/20                    | 1.0           | 547             | 0.460       | 0.384–0.527              | 12.6             | 4.0–20.2                      |
| 80/20                    | 2.0           | 547             | 0.479       | 0.401–0.550              | 15.9             | 8.5–22.6                      |
| 80/20                    | 3.0           | 547             | 0.536       | 0.469–0.600              | 19.4             | 13.6–24.9                     |
| 80/20                    | 5.0           | 547             | 0.570       | 0.503–0.630              | 21.3             | 14.7–27.3                     |
| 90/10                    | 1.0           | 355             | 0.327       | 0.218–0.436              | 12.8             | 4.4–20.3                      |
| 90/10                    | 2.0           | 355             | 0.358       | 0.251–0.457              | 13.0             | 5.3–20.3                      |
| 90/10                    | 3.0           | 355             | 0.416       | 0.316–0.502              | 14.5             | 6.7–21.8                      |

The site-and-regional-parameter model is shown. In the reverse 80/20 test at 3.0 s, KiK-net-to-K-NET estimation gives a correlation of 0.302 and an RMSE gain of  $-3.3\%$  (95% interval  $-7.8\%$  to  $1.5\%$ ).

**Supplementary Table 21.** European-Mediterranean repeatability and cross-region estimation

| Analysis                      | Period (s) | Records | Earthquakes | Mean paired stations/comparison | Group correlation (95% interval) | RMSE gain % (95% interval) |
|-------------------------------|------------|---------|-------------|---------------------------------|----------------------------------|----------------------------|
| Bindi post-2015               | 1.0        | 13,430  | 634         | 173.8                           | 0.929 (0.899–0.950)              | 66.8 (60.5–71.6)           |
| Bindi post-2015               | 2.0        | 13,430  | 634         | 173.8                           | 0.939 (0.915–0.957)              | 69.5 (64.7–73.6)           |
| Bindi post-2015               | 3.0        | 13,430  | 634         | 173.8                           | 0.928 (0.896–0.951)              | 66.4 (60.9–70.9)           |
| Bindi post-2015, $M_w \geq 5$ | 1.0        | 5,411   | 213         | 63.4                            | 0.878 (0.822–0.912)              | 53.4 (43.8–60.0)           |
| Bindi post-2015, $M_w \geq 5$ | 2.0        | 5,411   | 213         | 63.4                            | 0.875 (0.828–0.910)              | 52.8 (44.7–59.4)           |
| Bindi post-2015, $M_w \geq 5$ | 3.0        | 5,411   | 213         | 63.4                            | 0.871 (0.818–0.908)              | 52.2 (43.2–59.4)           |
| Kotha RotD50                  | 1.0        | 1,616   | 72          | 23.2                            | 0.860 (0.498–0.934)              | 48.8 (19.5–62.9)           |
| Kotha RotD50                  | 2.0        | 1,616   | 72          | 23.2                            | 0.916 (0.507–0.945)              | 58.1 (42.2–64.6)           |
| Kotha RotD50                  | 3.0        | 1,616   | 72          | 23.2                            | 0.919 (0.531–0.953)              | 58.5 (–4.2–66.1)           |

| Period (s) | Japanese estimation stations | ESM target stations | Correlation | Correlation 95% interval | RMSE gain % (95% interval) |
|------------|------------------------------|---------------------|-------------|--------------------------|----------------------------|
| 1.0        | 1,039                        | 265                 | 0.079       | –0.027–0.192             | –5.9 (–12.3––0.8)          |
| 2.0        | 1,039                        | 265                 | 0.043       | –0.072–0.163             | –3.7 (–8.0–0.1)            |
| 3.0        | 1,039                        | 265                 | 0.047       | –0.063–0.157             | –2.8 (–6.6–0.6)            |

The first block reports means across five earthquake-group comparisons and station-cluster bootstrap intervals. The Kotha 3.0 s RMSE interval crosses zero and is treated as a sample-limited sensitivity. The second block applies the fixed Japanese K-NET relation with  $\log_{10}(V_{S30})$  and elevation to ESM without refitting; its target-estimate mean is removed without using ESM station terms.

**Supplementary Table 22.** Local ESM spatial and earthquake-space estimation

| Validation design           | Model               | Period (s) | Stations | Correlation (95% interval) | RMSE gain % (95% interval) |
|-----------------------------|---------------------|------------|----------|----------------------------|----------------------------|
| Spatial block               | VS30 + elevation    | 1.0        | 236      | 0.106 (–0.008–0.217)       | –2.6 (–8.5–2.2)            |
| Spatial block               | Metadata            | 1.0        | 236      | 0.078 (–0.034–0.188)       | –4.7 (–11.0–0.6)           |
| Spatial block               | Metadata + location | 1.0        | 236      | –0.106 (–0.225–0.022)      | –10.4 (–15.6––5.5)         |
| Spatial block               | VS30 + elevation    | 2.0        | 236      | 0.118 (0.000–0.237)        | –2.6 (–8.1–2.1)            |
| Spatial block               | Metadata            | 2.0        | 236      | 0.115 (0.000–0.226)        | –3.6 (–9.8–1.5)            |
| Spatial block               | Metadata + location | 2.0        | 236      | 0.015 (–0.102–0.137)       | –7.7 (–13.3––2.6)          |
| Spatial block               | VS30 + elevation    | 3.0        | 236      | 0.074 (–0.043–0.189)       | –3.6 (–9.8–1.3)            |
| Spatial block               | Metadata            | 3.0        | 236      | 0.052 (–0.061–0.163)       | –5.5 (–12.8––0.03)         |
| Spatial block               | Metadata + location | 3.0        | 236      | 0.017 (–0.090–0.131)       | –7.6 (–13.1––2.2)          |
| Fifth group + spatial block | VS30 + elevation    | 1.0        | 207      | –0.004 (–0.123–0.113)      | –3.3 (–8.0–0.5)            |
| Fifth group + spatial block | Metadata            | 1.0        | 207      | –0.024 (–0.139–0.087)      | –5.7 (–10.9––1.9)          |
| Fifth group + spatial block | Metadata + location | 1.0        | 207      | –0.107 (–0.213–0.000)      | –13.9 (–21.3––8.4)         |
| Fifth group + spatial block | VS30 + elevation    | 2.0        | 207      | 0.043 (–0.065–0.153)       | –3.5 (–8.5–0.7)            |
| Fifth group + spatial block | Metadata            | 2.0        | 207      | 0.025 (–0.082–0.133)       | –5.8 (–11.3––1.4)          |
| Fifth group + spatial block | Metadata + location | 2.0        | 207      | –0.029 (–0.133–0.068)      | –8.3 (–14.3––3.9)          |
| Fifth group + spatial block | VS30 + elevation    | 3.0        | 207      | 0.021 (–0.083–0.132)       | –3.7 (–8.6–0.5)            |
| Fifth group + spatial block | Metadata            | 3.0        | 207      | 0.001 (–0.095–0.097)       | –6.4 (–11.9––2.3)          |
| Fifth group + spatial block | Metadata + location | 3.0        | 207      | 0.006 (–0.094–0.102)       | –5.4 (–10.2––1.6)          |

RMSE gain is relative to a zero station term; negative values indicate higher estimation error. Metadata adds slope and ESM housing, installation and reference-site fields. Location adds longitude and latitude. Intervals are from 2,000 station-cluster bootstrap replicates. The fifth-group design reports the aggregate across five earthquake-group comparisons.

**Supplementary Table 23.** J-SHIS source-category SA(3.0 s) sensitivity at the 50-year 10% level

| Product         | Stations | J-SHIS reference | Surface median SA (g) | Adjusted | Larger-category count |
|-----------------|----------|------------------|-----------------------|----------|-----------------------|
| All earthquakes | 1,628    | 0.078            | 0.081                 | 0.075    | –                     |
| Active shallow  | 1,628    | 0.031            | 0.032                 | 0.029    | 200                   |
| Subduction      | 1,628    | 0.071            | 0.072                 | 0.067    | 1,428                 |

J-SHIS spectral accelerations use the  $V_S = 400 \text{ m s}^{-1}$  engineering-bedrock reference. Surface values apply the MF2013 AVS30 projection; adjusted values also apply the cross-validated station terms. Category counts compare active-shallow and subduction values station by station. Fixed-probability category values are not additive.

**Supplementary Table 24.** Source-conditioned spatial estimation at long periods

| Source class | Period (s) | Stations | Correlation (95% interval) | RMSE gain % (95% interval) |
|--------------|------------|----------|----------------------------|----------------------------|
| Crustal      | 1.0        | 1,084    | 0.484 (0.438–0.530)        | 12.7 (8.2–17.2)            |
| Crustal      | 2.0        | 1,084    | 0.468 (0.414–0.517)        | 10.5 (5.8–14.8)            |
| Crustal      | 3.0        | 1,084    | 0.455 (0.403–0.507)        | 8.9 (3.9–13.8)             |
| Crustal      | 5.0        | 1,084    | 0.383 (0.329–0.437)        | 4.2 (–0.7–8.9)             |
| Interplate   | 1.0        | 779      | 0.384 (0.323–0.444)        | 4.5 (–1.1–9.9)             |
| Interplate   | 2.0        | 779      | 0.228 (0.153–0.301)        | –0.8 (–6.2–4.5)            |
| Interplate   | 3.0        | 779      | 0.251 (0.175–0.322)        | 2.8 (–1.7–7.4)             |
| Interplate   | 5.0        | 779      | 0.298 (0.228–0.360)        | 1.7 (–2.8–5.9)             |
| Intraplate   | 1.0        | 856      | 0.404 (0.345–0.461)        | 4.6 (–0.1–9.1)             |
| Intraplate   | 2.0        | 856      | 0.396 (0.332–0.453)        | 7.4 (3.0–11.6)             |
| Intraplate   | 3.0        | 856      | 0.516 (0.460–0.567)        | 14.3 (10.1–18.3)           |
| Intraplate   | 5.0        | 856      | 0.548 (0.502–0.594)        | 14.8 (10.7–18.6)           |

The fixed site-and-regional-parameter model is shown. Correlations compare record-derived and cross-validated station terms. RMSE gain is measured against a zero source-conditioned station term. Intervals use 2,000 station-cluster bootstrap replicates. Results at every direct period are retained in the accompanying CSV.

**Supplementary Table 25.** Kanto SA(5.0 s) comparison with the Sung et al. simulation field

| Japanese station terms      | Stations | Pearson (95% interval) | Spearman (95% interval) | Sign agreement % (95% interval) |
|-----------------------------|----------|------------------------|-------------------------|---------------------------------|
| Crustal, record-derived     | 364      | 0.304 (0.163–0.434)    | 0.372 (0.227–0.494)     | 62.9 (56.5–69.5)                |
| Crustal, site + regional    | 364      | 0.135 (0.038–0.231)    | 0.141 (0.005–0.276)     | 54.9 (49.1–60.3)                |
| Crustal, site parameters    | 364      | 0.214 (0.123–0.305)    | 0.233 (0.119–0.348)     | 57.4 (50.9–63.1)                |
| All-source, record-derived  | 389      | 0.122 (–0.009–0.258)   | 0.173 (0.021–0.332)     | 55.8 (48.2–62.2)                |
| All-source, site + regional | 389      | 0.349 (0.243–0.446)    | 0.407 (0.280–0.513)     | 63.5 (57.8–69.1)                |
| All-source, site parameters | 389      | 0.298 (0.196–0.395)    | 0.326 (0.194–0.442)     | 57.8 (52.2–64.9)                |

Cross-validated estimates are calculated outside each station’s spatial block. Both patterns are expressed in natural-log units and centred over the matched stations. Intervals use 2,000 resamples of 31.7 km spatial blocks. Record-derived crustal station terms provide the prespecified source-matched comparison.

**Supplementary Table 26.** Source-conditioned J-SHIS SA(3.0 s) sensitivity at the 50-year 10% level

| Product / station terms  | Stations | J-SHIS reference median SA (g) | Surface | Adjusted | Multiplier 5th–95th percentile |
|--------------------------|----------|--------------------------------|---------|----------|--------------------------------|
| Active shallow / crustal | 1,084    | 0.033                          | 0.034   | 0.033    | 0.642–1.517                    |
| Subduction / interplate  | 779      | 0.077                          | 0.082   | 0.084    | 0.711–1.430                    |
| Subduction / intraplate  | 856      | 0.082                          | 0.089   | 0.083    | 0.661–1.416                    |

Each row uses the stations supported by its source-conditioned terms. Interplate and intraplate rows are separate sensitivities on the J-SHIS subduction product. A source-level PSHA recomputation requires source recurrence, source weights and probability aggregation that are absent from the fixed-probability maps.

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